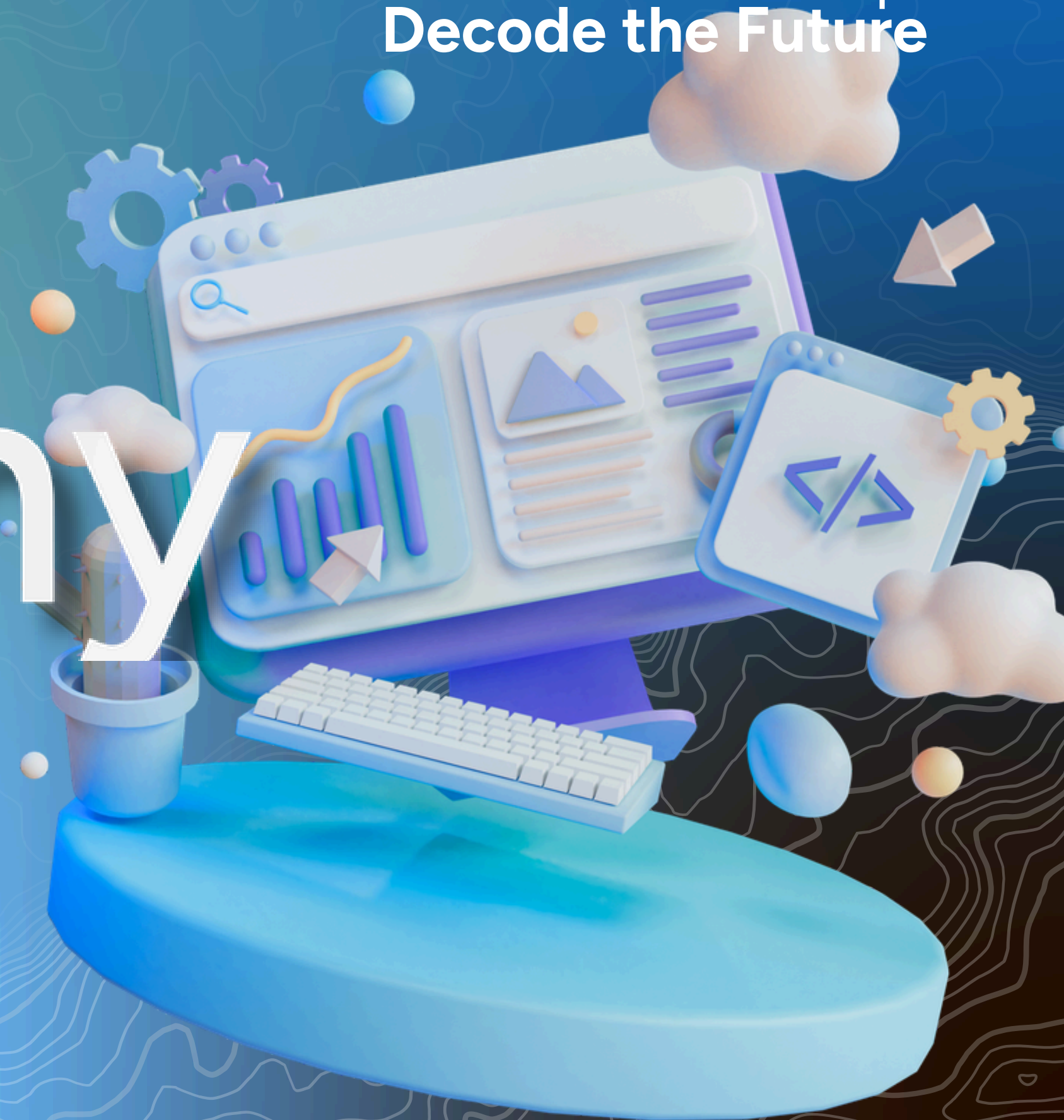


2026

Code the Map
Decode the Future

MAPID Academy

WebGIS Development Bootcamp



Outline

Location Value

Materials:

1. Konsep Perspektif Lokasi
2. Problem Definition & Spatial Thinking
3. Location Analytics (Business Cases)
4. Introduction to GEO MAPID (Map Viewer & Map Editor)
5. GEO MAPID as Database

Hands-On:

1. Project Definition Trial
2. Understanding Basic Features GEO MAPID
3. Quickstart Mapping on GEO MAPID

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03

Location Value

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Lokasi menjadi salah satu aspek yang punya value dalam mengidentifikasi kebutuhan bisnis.

**Masayoshi Son**
Founder, Chairman & CEO

SoftBank was founded for what purpose? For what purpose was Masa Son born? It may sound strange, but I think I was born to realize ASI. I am super serious about it. Are you serious, you may ask, can you do it? Actually, that was a question I asked myself. But this morning, I solved the most complex problem. That's why I am super confident today. Don't ask me why. Looking back, evolution of humanity, 200,000 years back, human used to drive evolution. Humans created tools, humans drove evolution. Pretty soon, AGI will come, maybe in five years, even in three years, sometime in three years to five years, AGI will come. Beyond that, AGI will drive evolution. In the past, geniuses have driven evolution of humanity, different geniuses stimulated each other and delivered evolution. Going forward, AGI stimulates AGI and drives evolution. Because AGI is much smarter than humans. Smarter AGIs stimulate smarter AGIs accelerate evolution. It's non-stoppable. Then, ASI will come. I think ASI will come sometime in 10 years. 10,000x smarter than humans. Looking back 200,000 years of the human history, compared to that, just less than 10 years, ASI will come.

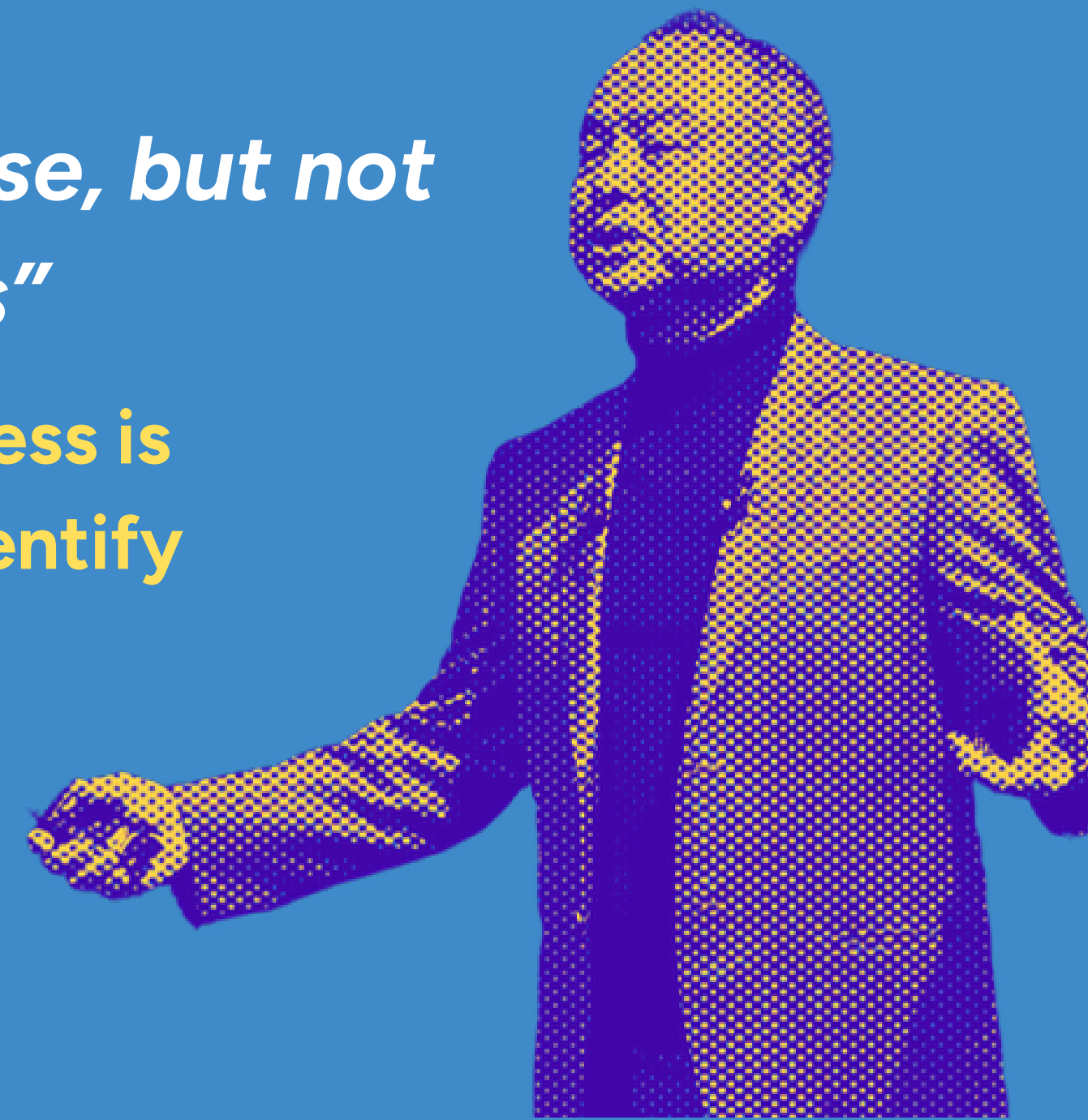
 SoftBank Group • AGM 2024 21 Jun, 2024

This highlight was created using [Quartr Pro](#)

“Every company has a purpose, but not all have meaningful purposes”

CTA: If the core purpose of a business is to create value, then how do we identify that specific value?

Let's think spatially!

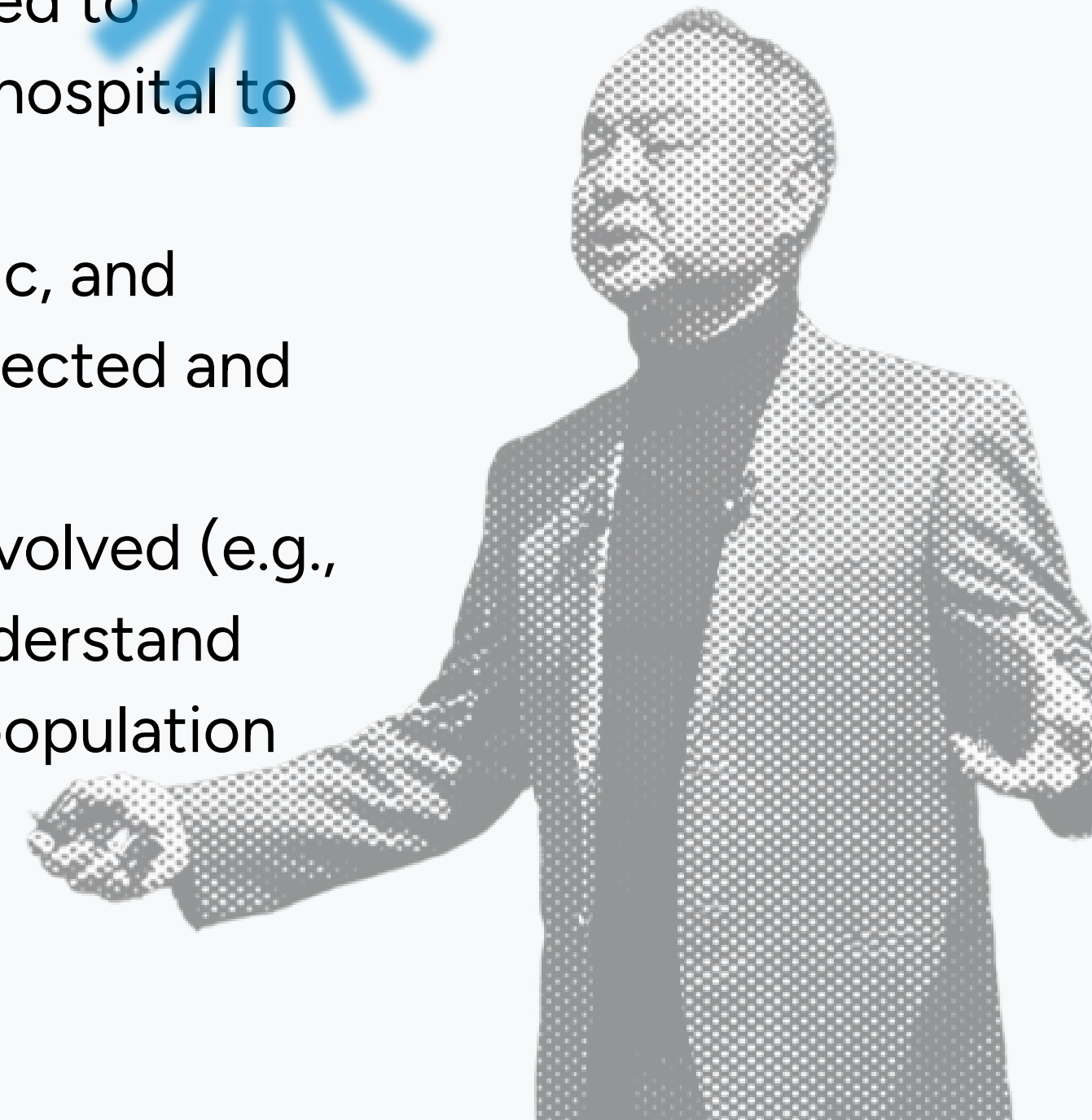


Understanding the Problems

Let's think spatially!



- **Define the Problem or Objective:** Clearly state what you need to address or achieve. Example: "Where should we build a new hospital to best serve the community?"
- **Understand the Context:** Consider physical, social, economic, and cultural aspects relevant to your problem. Identify who is affected and involved in the issue.
- **Identify Key Components:** List out the main components involved (e.g., population, transportation, existing healthcare facilities). Understand how these components interact with each other (e.g., how population density affects healthcare demand).





Edward Jones[®]
MAKING SENSE OF INVESTING

1980

Edward Jones conducted a series of analyses
and consultations

discovered that its resonating value proposition was that it
offered a highly personalized investment service to those
individual customers who wanted to delegate investment
decisions

rural community

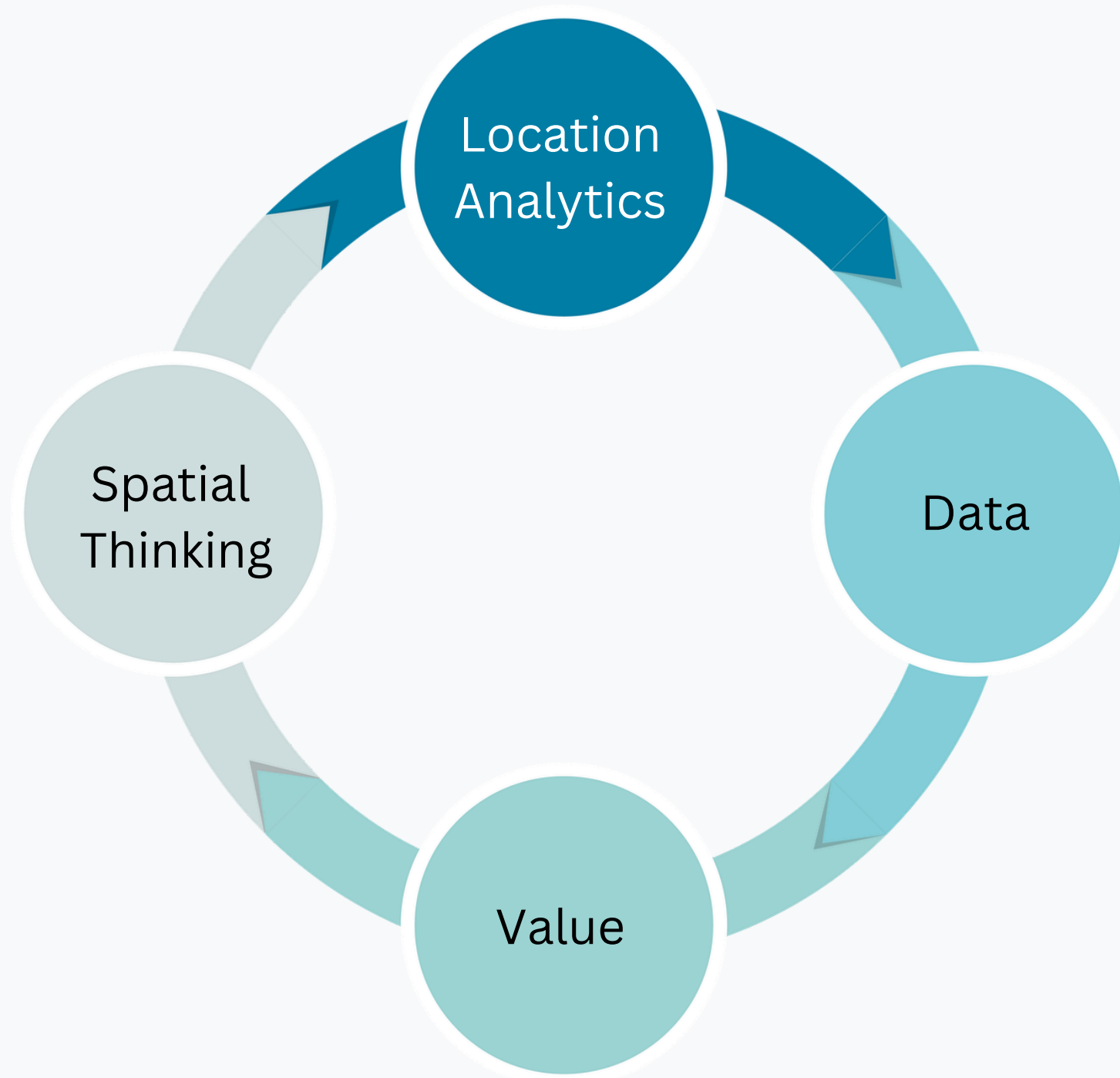


This spatial reasoning

resulted in the rapid growth of Edward Jones
from 400 to 1,000 locations in a seven-year period

Spatial Thinking/Reasoning

See This Cycle!



Value

Apa yang hendak dicapai oleh bisnis?

Spatial Thinking

Terjemahkan tujuan bisnis jadi pertanyaan tentang **LOKASI**

Location Analytics

Lakukan 4 teknik analisis, seperti **Deskriptif, Eksplanatori, Prediktif, dan Preskriptif**

Data

Perhatikan data lokasi yang akurat dan relevan

Catch the idea!

Value?

- Efficiently and quickly match passengers with drivers.
- Provide accurate and competitive estimates for time and pricing

Spatial Thinking?

Which driver is closest and has the fastest estimated time of arrival (ETA) to the passenger's location?

Location Analytics?

Data?



Location Analytics?

Code the Map
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Descriptive Analysis

What is happening?

Displaying a real-time map in your app that shows the locations of nearby drivers

Explanatory Analysis

Why is this happening?

Analyzing why wait times in Area X are consistently long every Friday night

Predictive Analysis

What is likely to happen?

Based on historical data and upcoming events (e.g., a concert at GBK), we predict a spike in goride at 10 PM.

Prescriptive Analysis

What should be done?

Based on historical data and upcoming events (e.g., a concert at GBK), we have to send notifications and offer incentives or bonuses to drivers near GBK before ends.

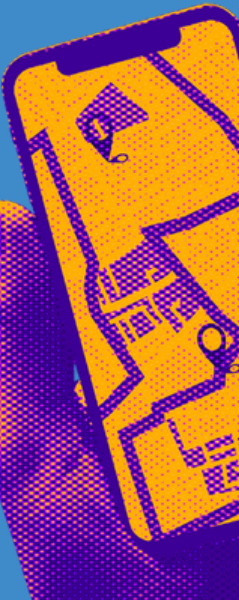
Data?

Real-time GPS data

Historical trip data

Third-party location data

Area polygons



Simple to Medium Processing

Digitasi Pada Layer

Code the Map
Decode the Future

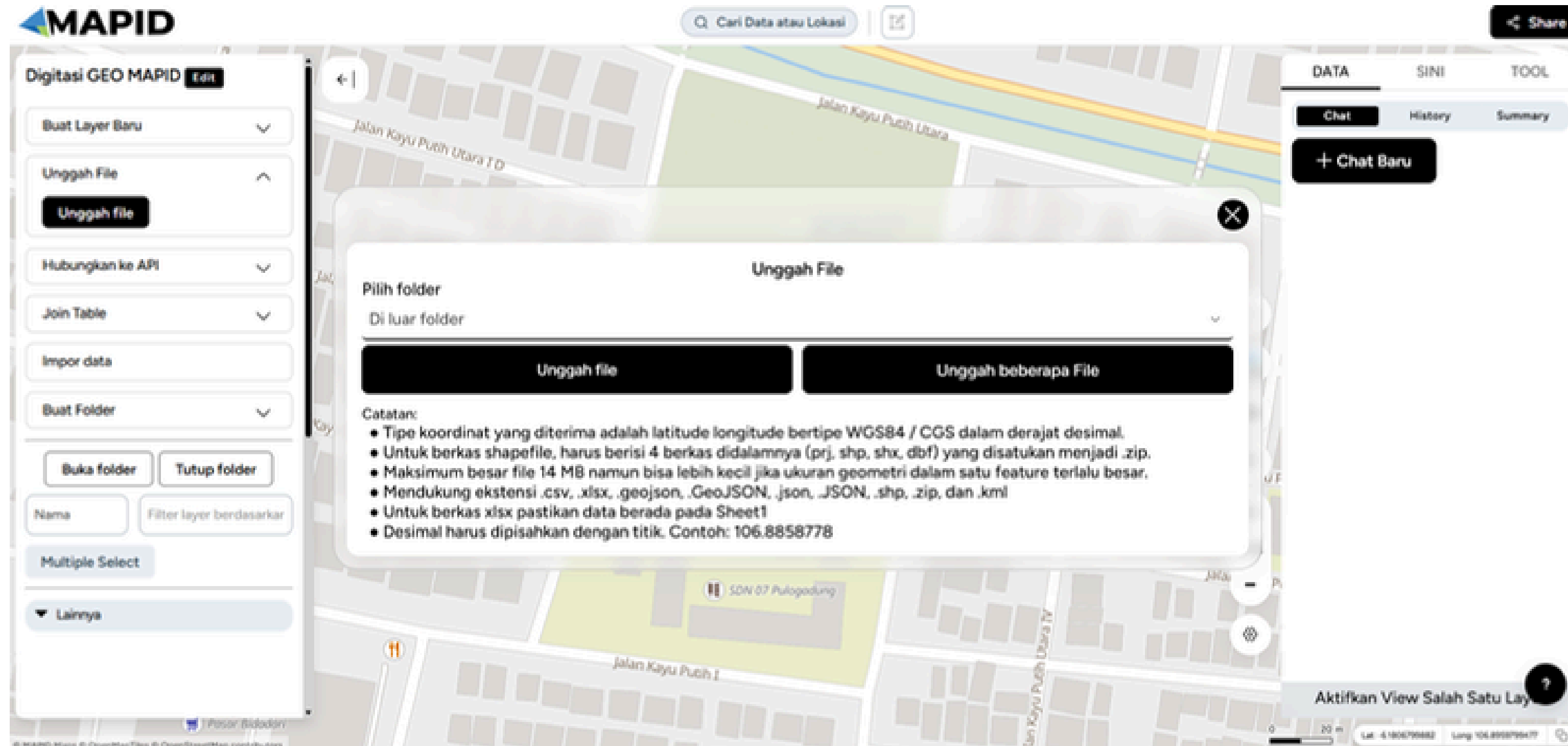
The screenshot displays the MAPID web application interface, which is used for digitizing map data. The interface is divided into several sections:

- Left Panel (Buat Layer Baru):** A sidebar for creating new layers. It includes fields for "Nama layer" and "Deskripsi layer", a "Point" dropdown menu, and a "Multiple Select" button. A red circle with the number 2 is placed next to this panel.
- Top Panel (Digitasi):** A header section with a search bar labeled "Cari Data atau Lokasi" and a red circle with the number 1. Below the search bar, there are buttons for "Buat Layer Baru", "Unggah File", "Hubungkan ke API", "Join Table", "Impor data", and "Buat Folder". A red circle with the number 4 is placed above the search bar.
- Map View:** A central map area showing a street grid. A blue polygon is drawn on the map, representing a digitized area. A red circle with the number 5 is placed over this polygon.
- Right Panel (DATA, SINI, TOOL):** A sidebar containing various tools and data management options. It includes a "Chat" button, a "History" button, a "Summary" button, and a "Bangunan" section with a description and project name. Below this, there are buttons for "Unduh Data" (XLSX, Geojson) and a "Pilih kolom untuk disaring" section. A red circle with the number 6 is placed next to the "Pilih kolom untuk disaring" section.
- Bottom Panel (Lainnya):** A bottom section showing a list of layers. A red circle with the number 3 is placed over the "Bangunan" layer, which is currently selected. The layer details show it was created by "Dzikri N J" and has a "Digitasi" status.

Simple to Medium Processing

Upload Data

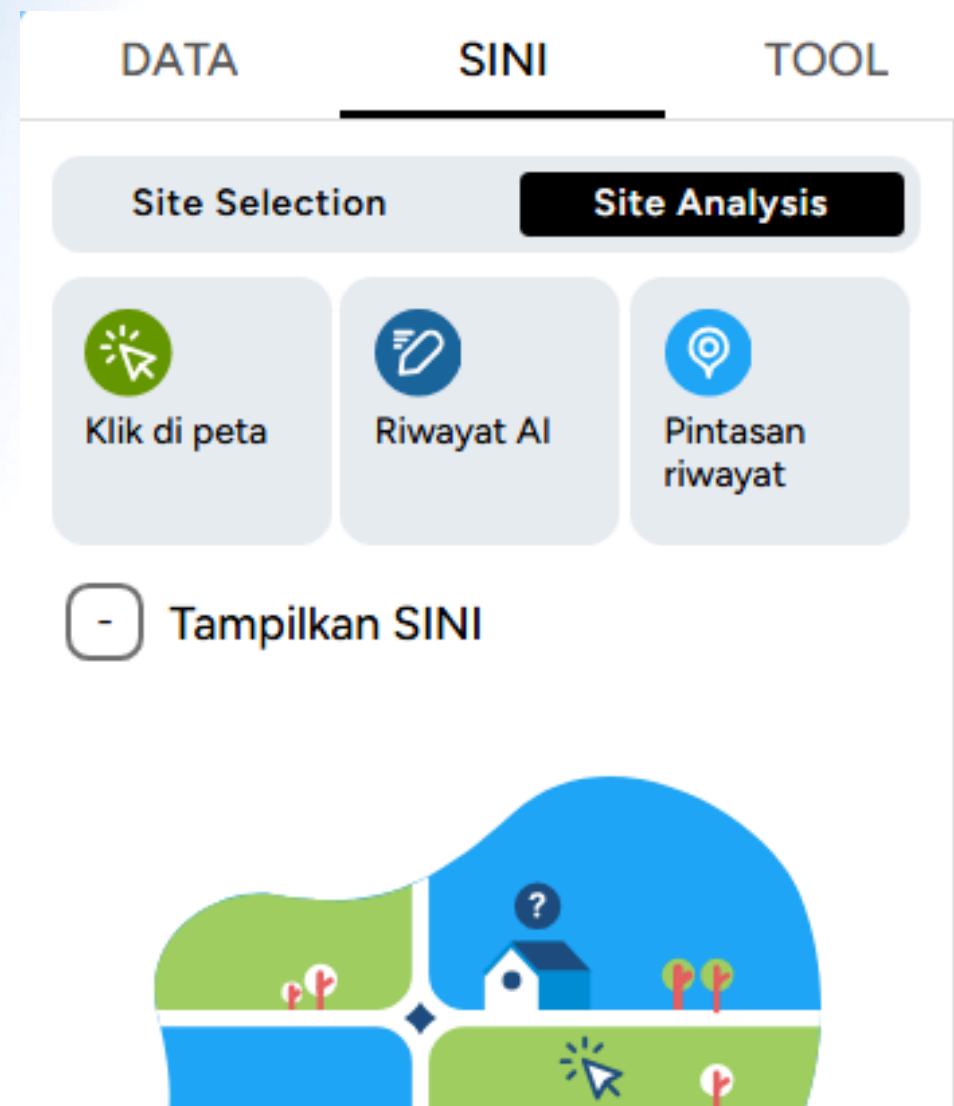
Code the Map
Decode the Future



GEO MAPID Tools



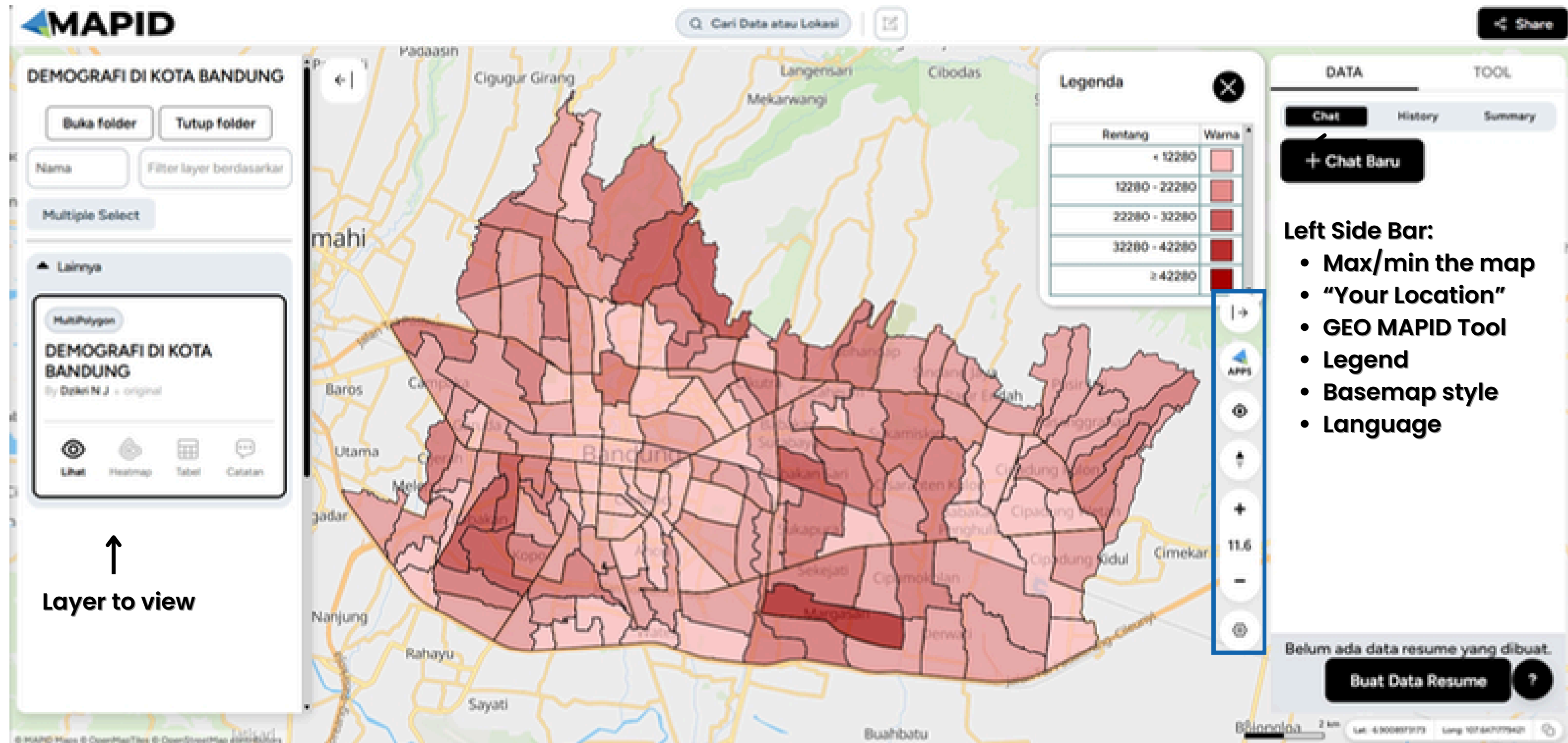
1



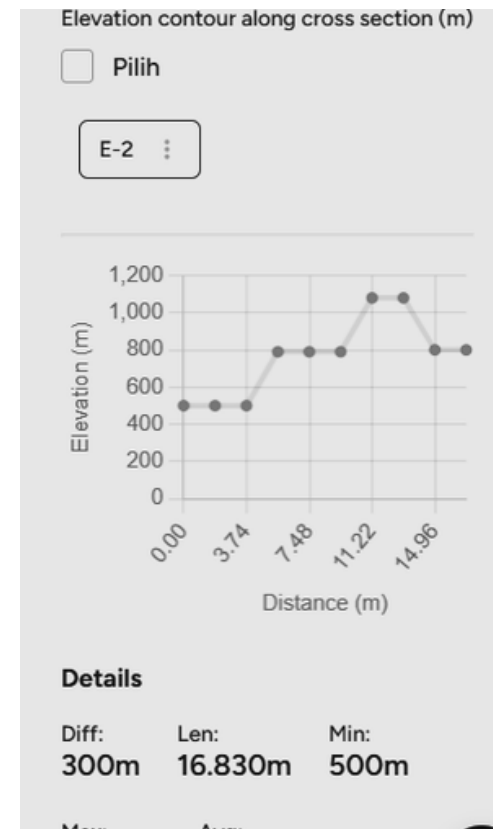
2



Map Viewer

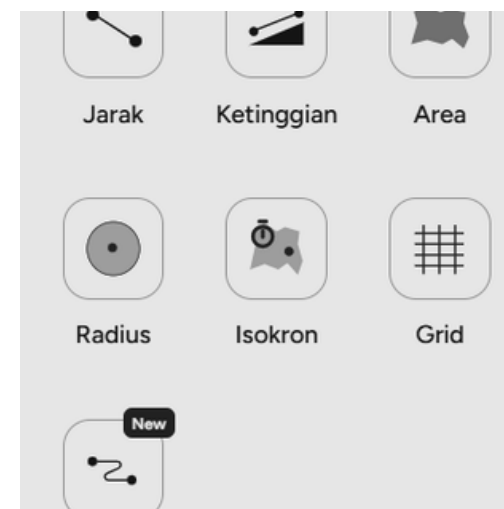
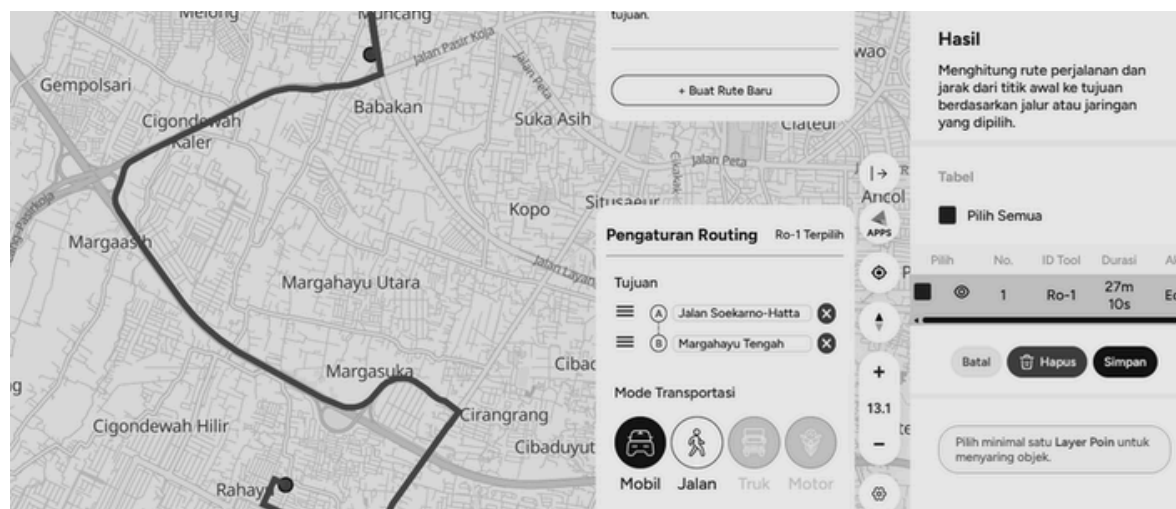


TOOLBOX



What can I do with TOOLBOX?

- Measure the distance or length of line features (linestrings)
- Calculate the area covered during travel
- Filter data based on a specified radius or area
- Create cross-sectional profiles along a defined line



Completed projects from MAPID

Transportation

Toll Price Database 2024

Property Appraisal

Retail Optimization

Site Selection for retail

**MAPID as a Solution
to Real-World
Problems**

Urban Planning

Strategic Locations for New
BRT Stops

Asset management

Tenant data collection
Logistics Optimization

Poor location decisions = business failure

Choosing the wrong location can lead to low customer traffic, high costs, and ultimately business collapse.

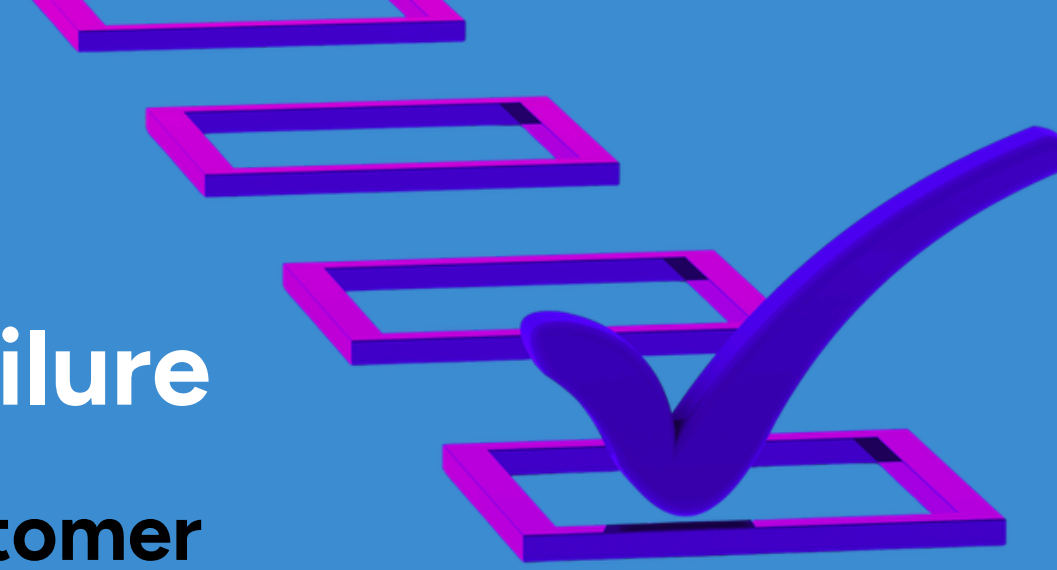
Gut-based decisions vs. Data-driven decisions

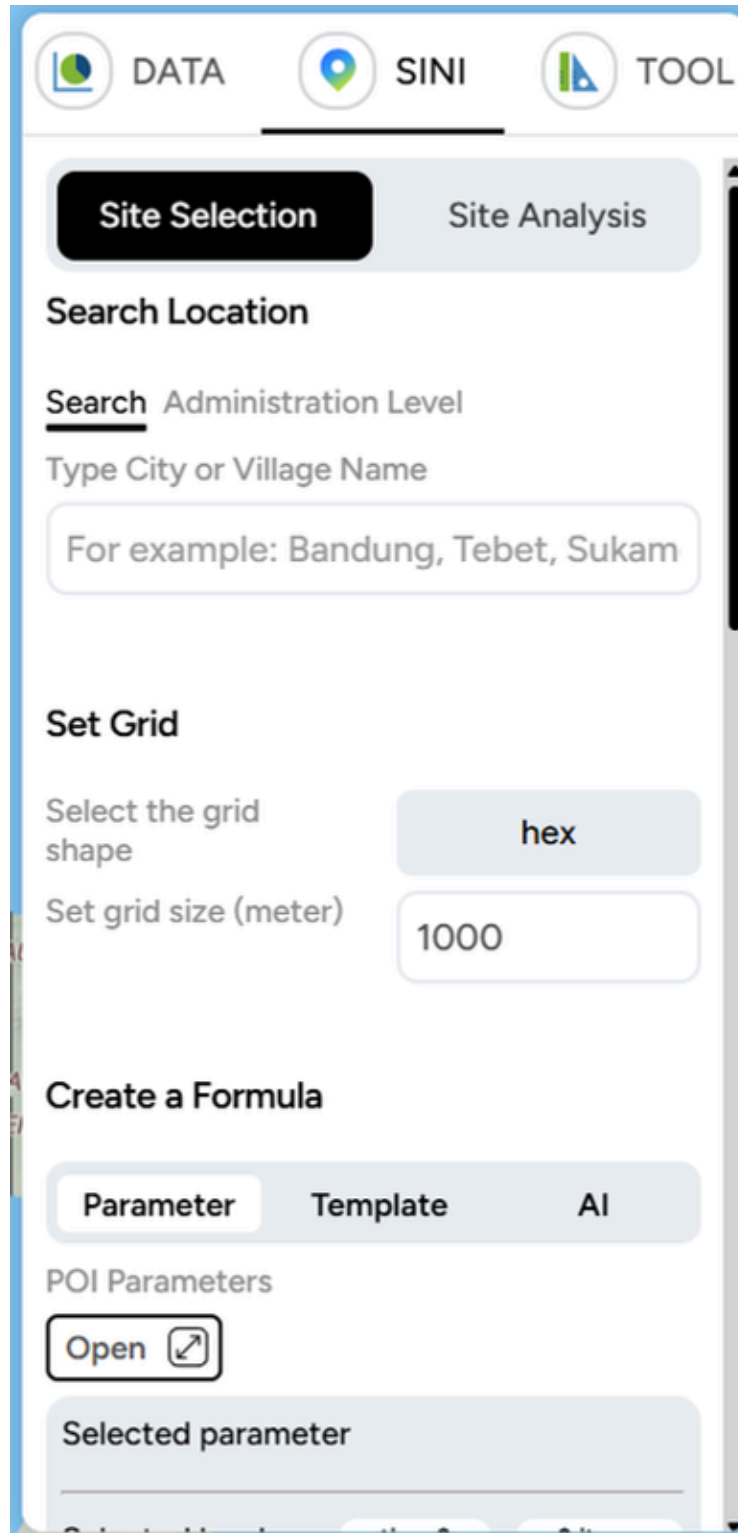
Intuition may feel right, but data-driven decisions are grounded in facts and reduce uncertainty.

Site selection is a systematic process to reduce risk and maximize potential

Uses structured analysis to find the best location, minimizing risk and boosting success.

Site Selection matters?





The screenshot shows a mobile application interface for 'SINI DATA: Site Selection'. At the top, there are three tabs: 'DATA', 'SINI', and 'TOOL'. Below these, there are two main sections: 'Site Selection' (active) and 'Site Analysis'. The 'Site Selection' section includes a 'Search Location' area with a 'Search' button and a dropdown for 'Administration Level'. Below this is a text input field for 'Type City or Village Name' with a placeholder example: 'For example: Bandung, Tebet, Sukam'. The 'Set Grid' section has a 'Select the grid shape' dropdown set to 'hex' and a 'Set grid size (meter)' input field set to '1000'. The 'Create a Formula' section has three tabs: 'Parameter', 'Template', and 'AI'. Below these is a 'POI Parameters' section with an 'Open' button and a 'Selected parameter' field.

SINI DATA: Site Selection

The process of selecting a location for development by considering multiple factors.

Identifying optimal locations in accordance with defined objectives

Supporting investment and development decision-making

Optimizing costs and benefits

The Framework: MCDA

Code the Map
Decode the Future

A Brief History

Started during World War II to solve simple problems like finding the fastest route. But, real-world are more complex

The Birth of MCDA

In the 1960s–70s, experts created MCDA to help people make better choices by comparing many factors clearly and logically.

Recipe We Use

An American School of MCDA (Weighted Scoring) to declare ingredients (important criteria) and measurements (weight of each)

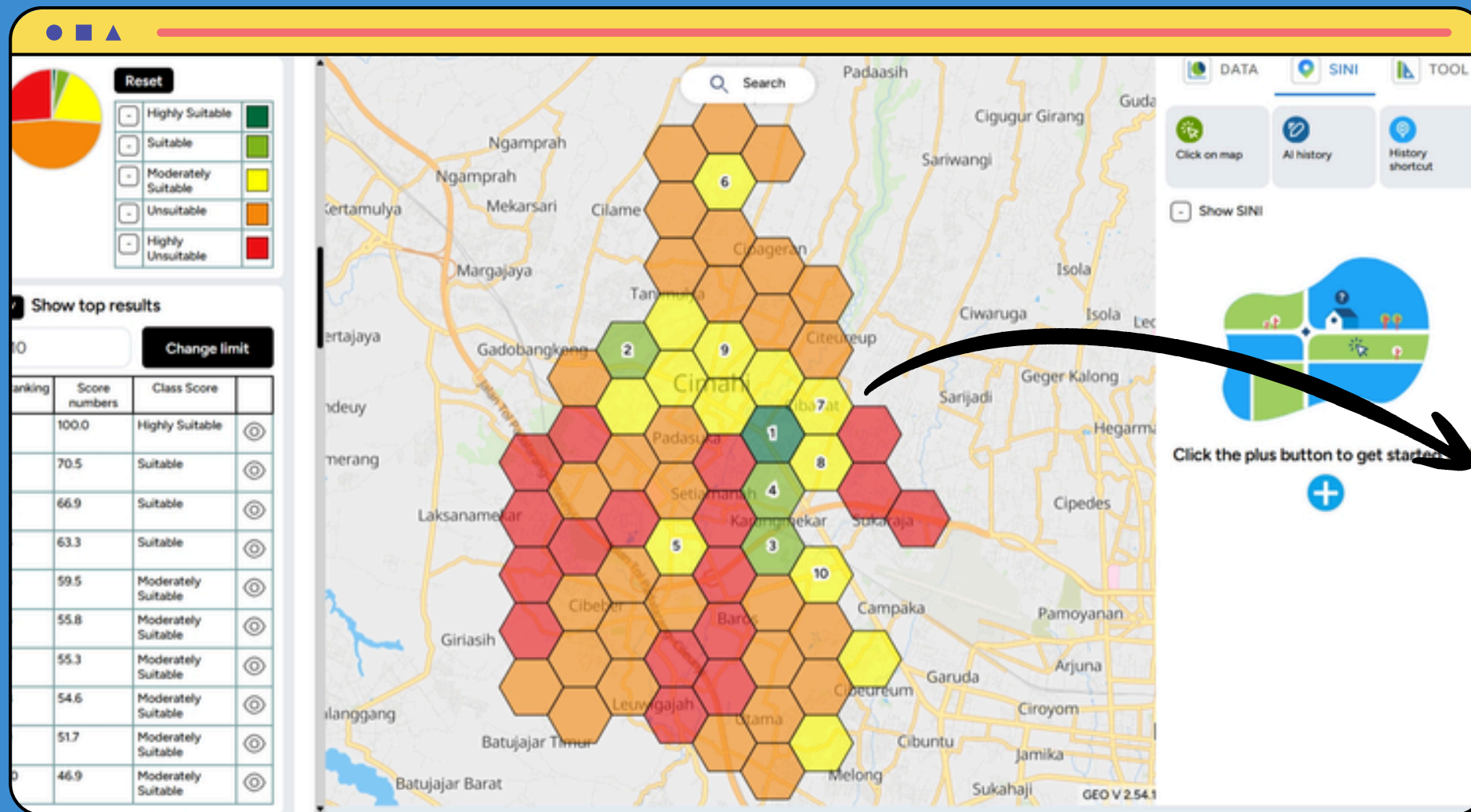
Macro*

VIEW

*After combining and weighting all the data, the platform creates a score map. The green areas are the best picks and will be checked more closely.



Code the Map
Decode the Future



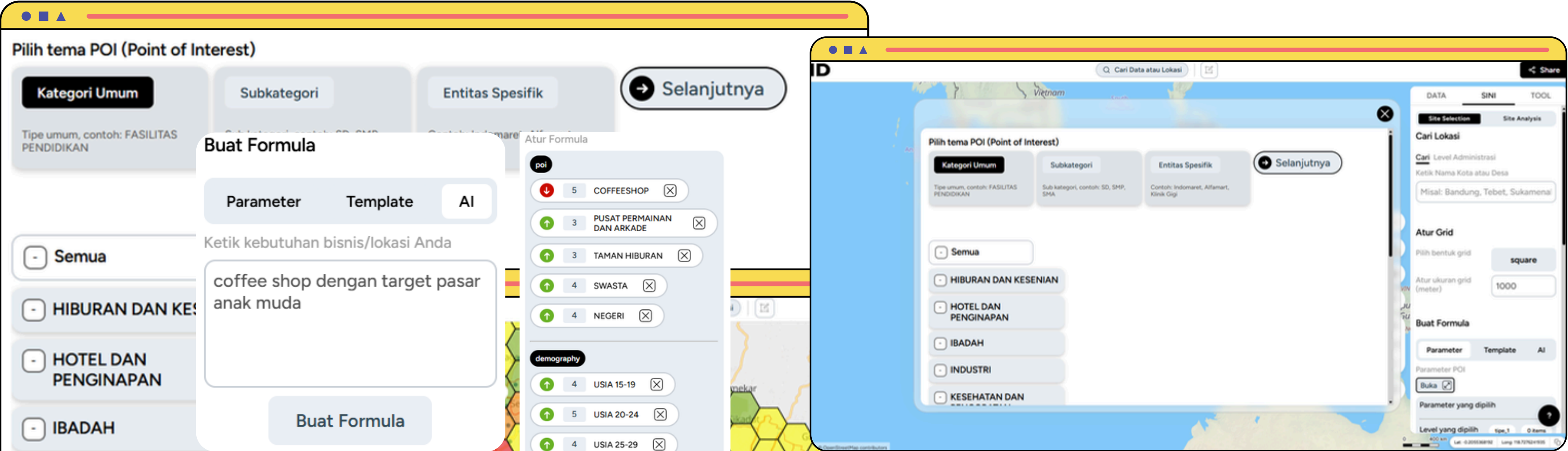
Dig deeper with 1/2/3/4 gold areas as you had calculated

Macro-level analysis helps identify the distribution of potential locations for deeper, hyperlocal analysis

Here are the yields

Insight (Site Selection)

Code the Map
Decode the Future



The Measurements

The Ingredients

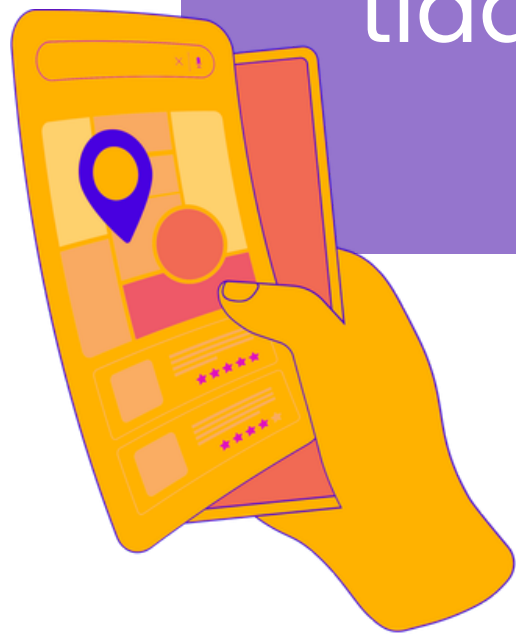
The Yield

After Macro-View

Try to respond!



Setelah kita mengidentifikasi *gold zones*, apa yang membuat analisis hiper-lokal menjadi langkah selanjutnya? Mengapa kita tidak bisa hanya mengandalkan data tingkat makro dalam mengambil keputusan lokasi akhir?



Why hyperlocal is significant?

Consumer demand and competition vary block by block. Macro-level analysis alone is no longer sufficient.

Every Location is a Unique Battlefield

Code the Map
Decode the Future

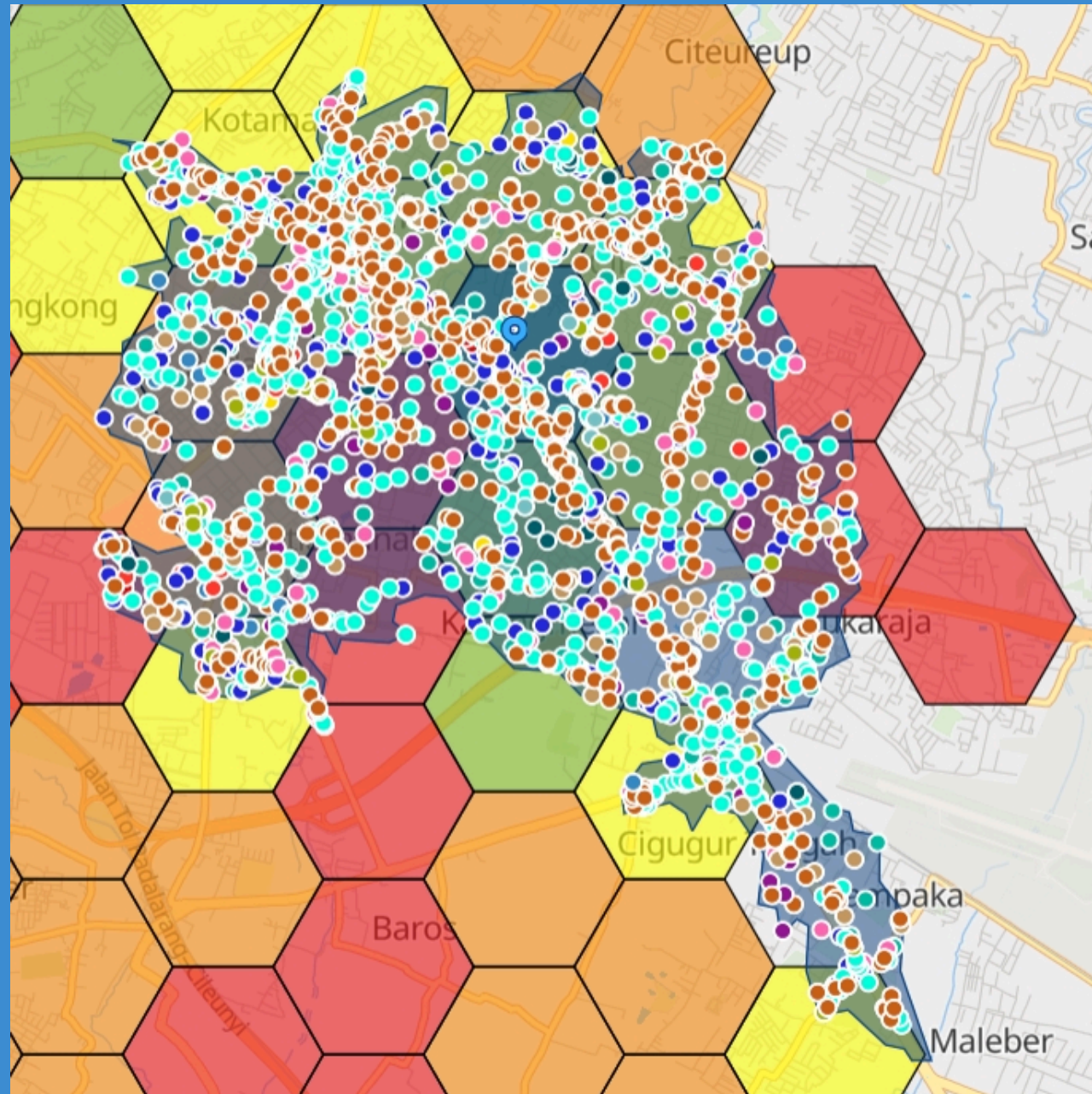
“in case of retail, with hyperlocal insight, it can optimize pricing, promotions, and product offerings to win at store level.”

AI as Your Brainstorming Partner

Retailers can significantly increase sales (2–4%) with an AI-based hyperlocal approach. Exp: for this high-SES area, consider proximity to premium fitness centers or international schools.



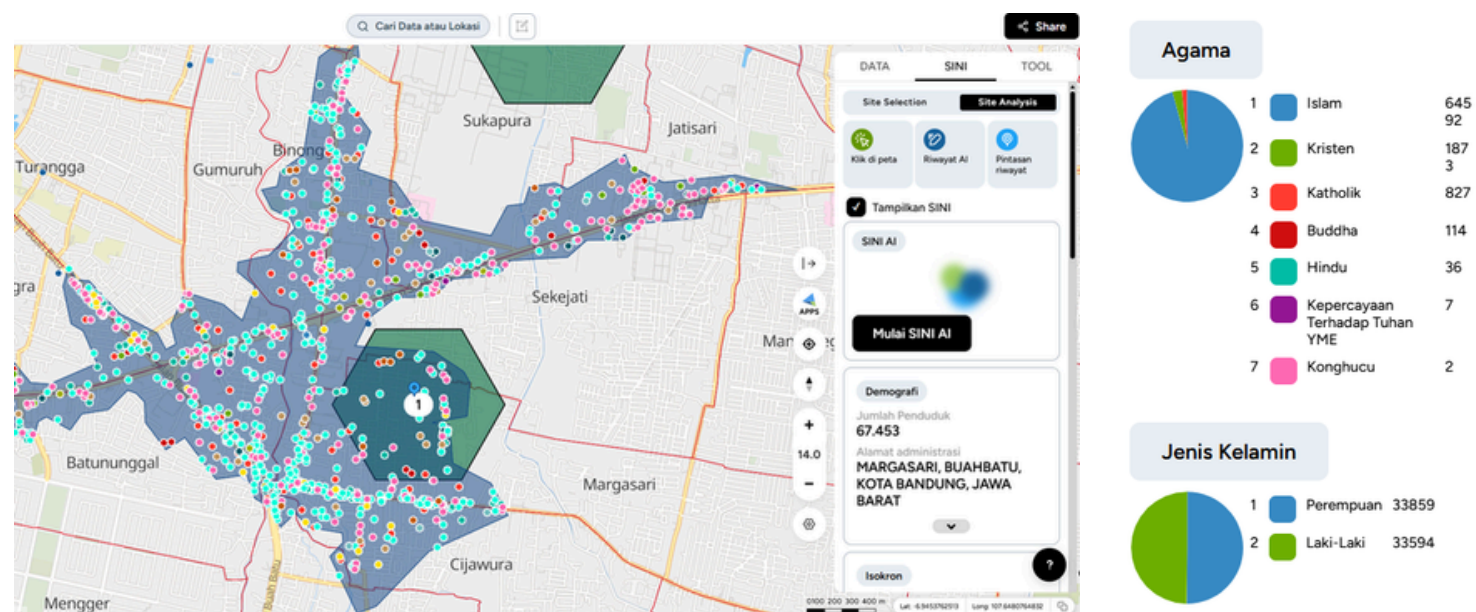
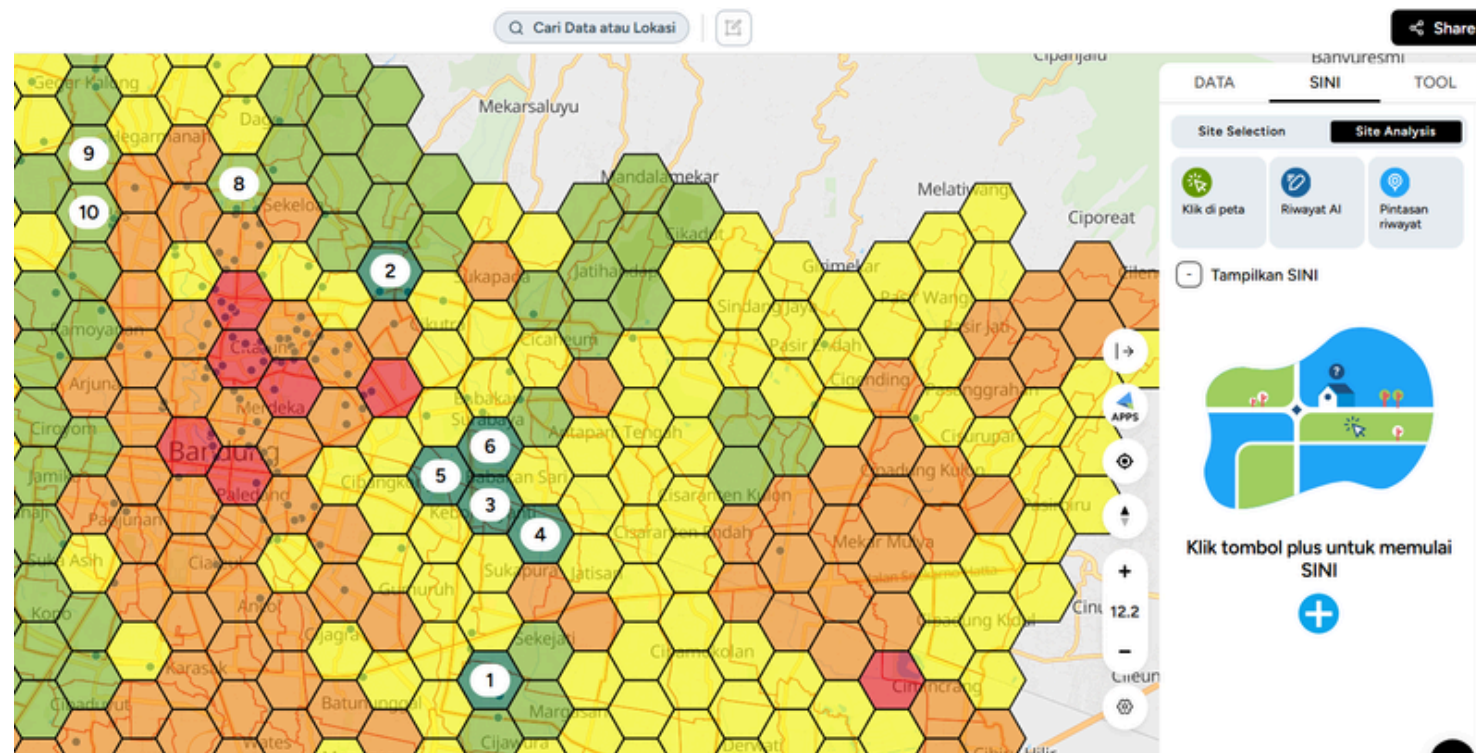
How to Win the Hyperlocal Sales Race, Retailers Must Turn to AI | Insights
How a 2-4% revenue uplift could be achievable with BCG's AI-driven Twin
BCG Platini / Apr 22, 2024



**Dig deeper with site analysis.
Look the opportunity of your
potential “point”! Consider the
demography, point of interest,
disaster, and etc**

Why hyperlocal is significant?

SINI & SINI AI



DATA SINI TOOL

di sini cocok bikin coffee shop?

Untuk mengetahui apakah suatu lokasi cocok untuk membuka coffee shop, ada beberapa faktor yang perlu dipertimbangkan, antara lain:

- 1. Populasi dan Kepadatan Penduduk:** Di wilayah Margasari, Buahbatu, Kota Bandung, jumlah penduduk pada tahun 2024 diperkirakan mencapai 67.453 jiwa dengan kepadatan penduduk sekitar 64.645 jiwa per km². Kepadatan penduduk yang tinggi dapat meningkatkan potensi pelanggan.
- 2. Daya Beli Masyarakat:** Pengeluaran rata-rata penduduk untuk makanan dan non-makanan juga dapat menjadi indikator penting. Di Kota Bandung, pengeluaran makanan rata-rata adalah Rp803.842,31 per kapita per bulan dan non-makanan Rp1.633.032,46 per kapita per bulan. Jika pengeluaran untuk minum kopi termasuk dalam pengeluaran bulanan ini, ada potensi pasar yang baik.

Tanya bisnis, harga tanah,

How powerful is SINI AI?

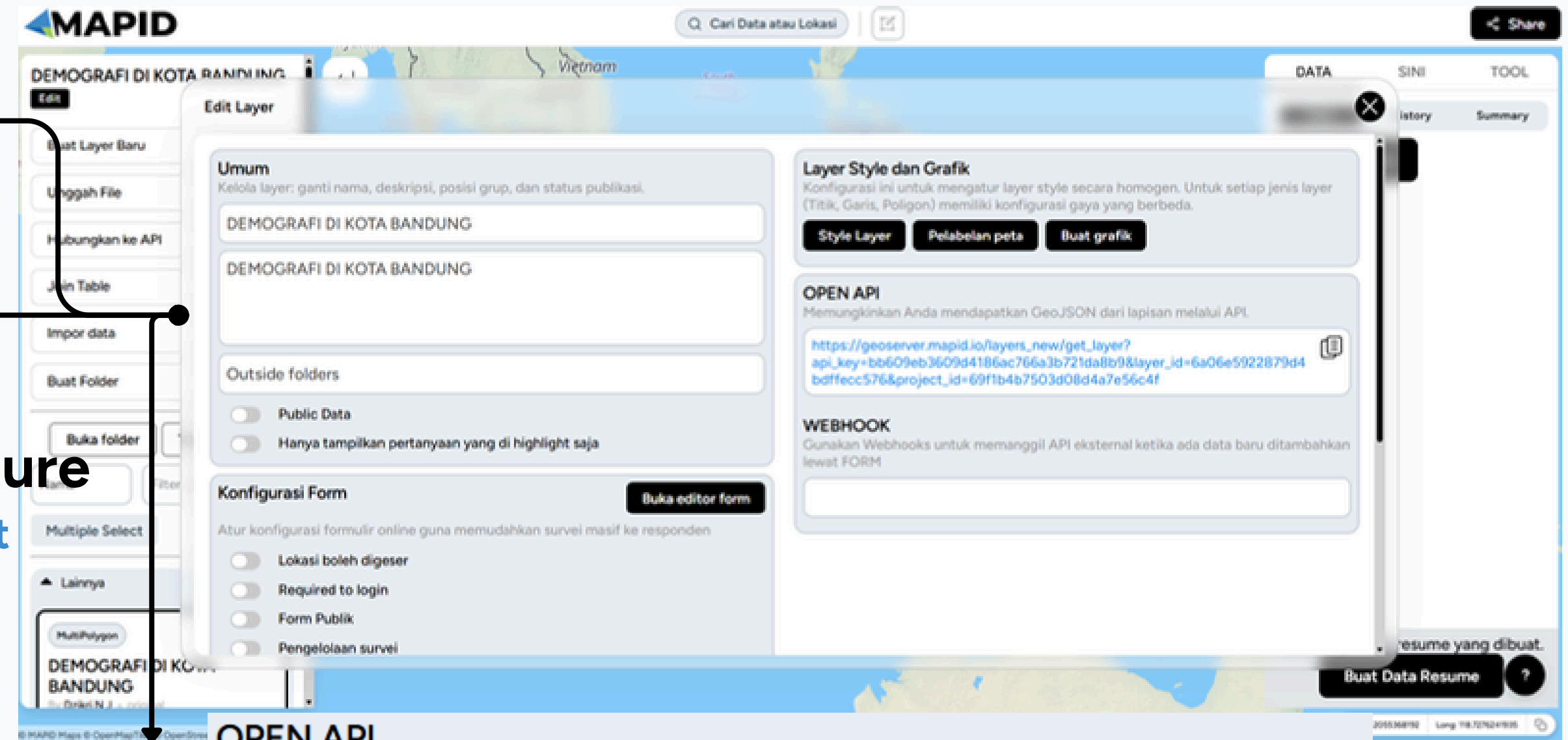
- Smart geo-based platform
- Access to all kinds of data
- Chat with AI like a buddy
- Cool data visuals and optimization
- Helps with location-based decisions

GEO MAPID as A Database

**Cloud-Based Spatial
Data Management Hub**
Organize and store your data in a
efficient manner

**Structured Layers for
Organized Data Architecture**
Data is arranged in layers to keep it
organized and easy to manage

**Dynamic Data Source for
WebGIS Integration (with API)**
Fetch data from GEO MAPID, process
with Python, and push the output back
to your custom WebGIS

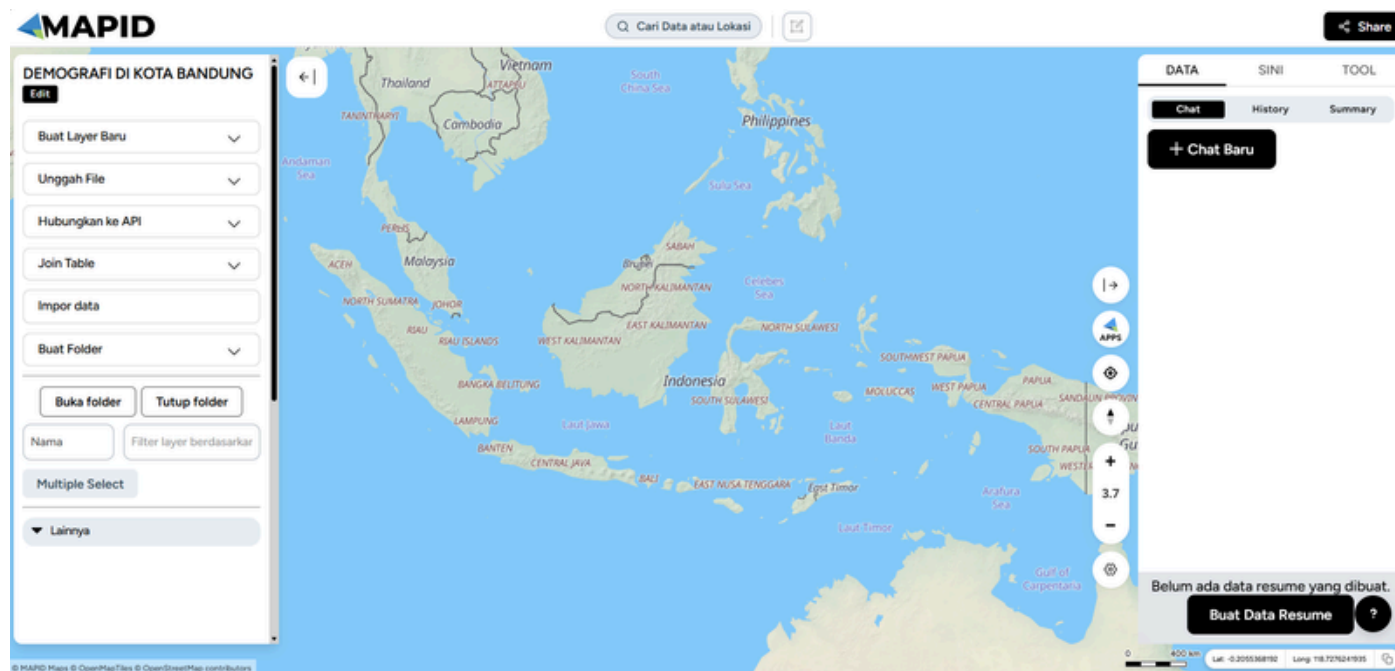


OPEN API

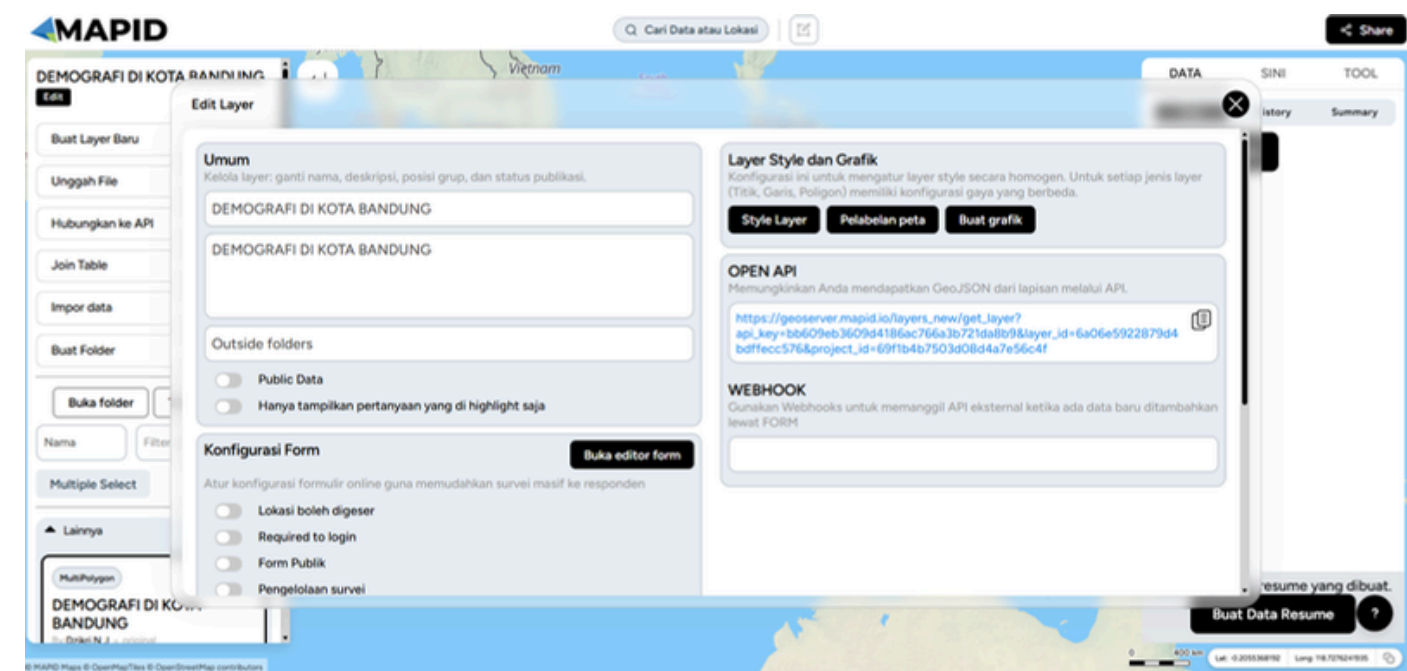
Memungkinkan Anda mendapatkan GeoJSON dari lapisan melalui API.

https://geoserver.mapid.io/layers_new/get_layer?api_key=bb609eb3609d4186ac766a3b721da8b9&layer_id=6a06e5922879d4bdffecc576&project_id=69f1b4b7503d08d4a7e56c4f

How to Access Open API Layer



Pilih **Project** Tempat Anda Menyimpan Data



Pilih **Layer** untuk Diakses untuk data WebGISmu

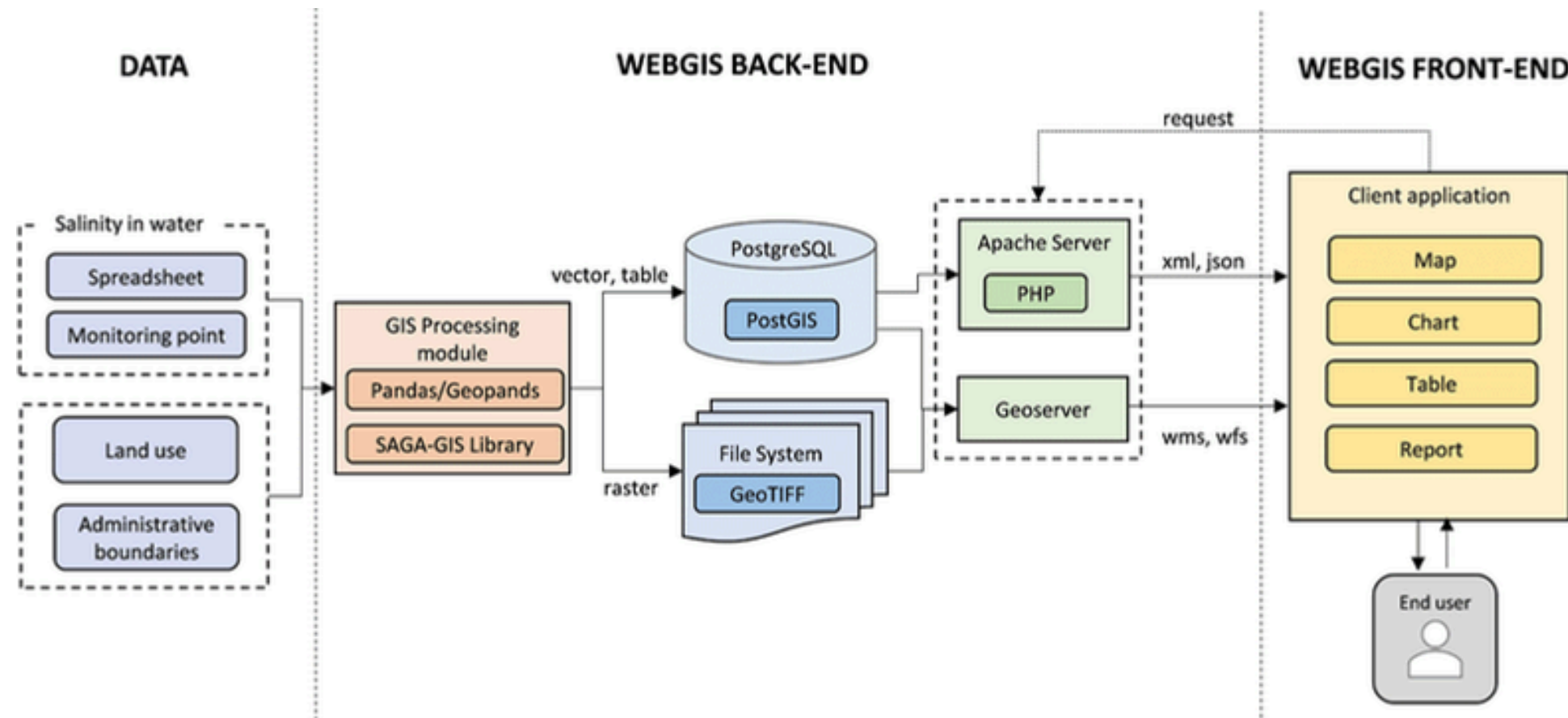


Access via your code!

WebGIS Fundamental

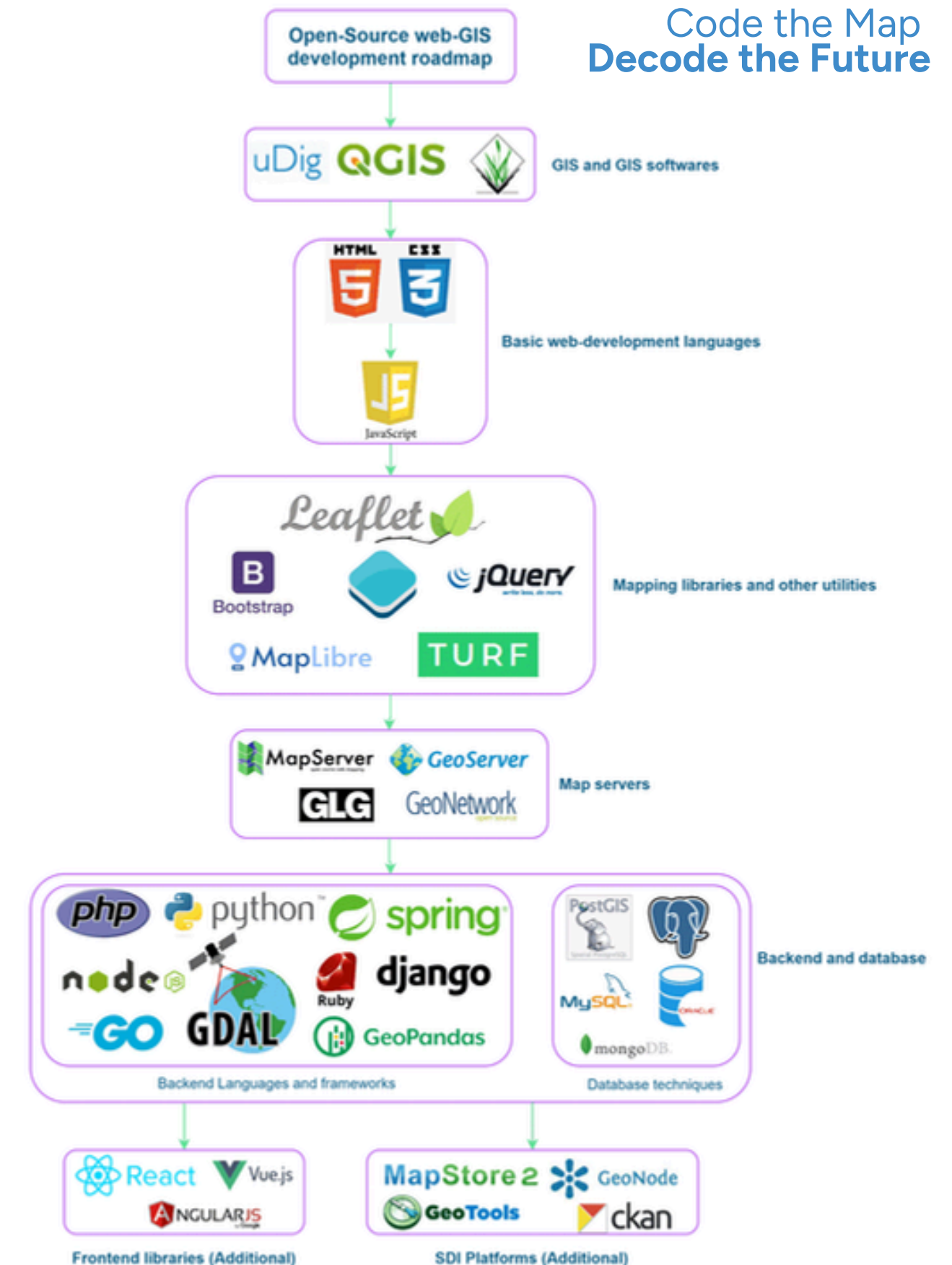
From Desktop-Based to Connected Future

 Summary



Binh Tran Thai et al, 2020

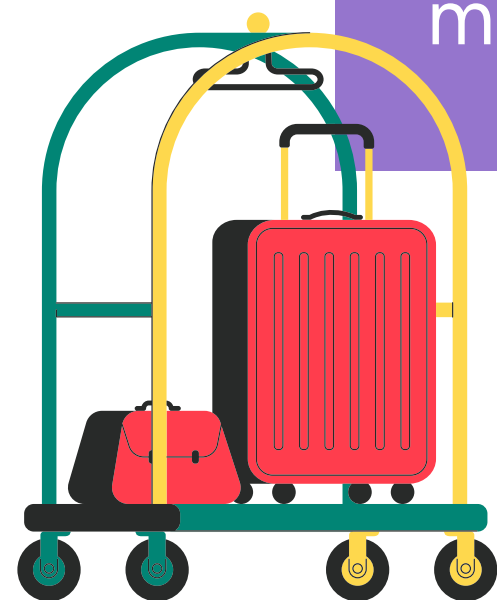
Next Journey: Web Fundamental



Copyright:
Tek Kshetri and Emmanuel Jolaiya

Anda diminta untuk mengeksplorasi bagaimana data di GEO MAPID dapat digunakan untuk memahami peluang lokasi baru. Fokusnya bukan sekadar memilih titik, tetapi melihat bagaimana transportasi, POI pendukung, demografi, dan infrastruktur saling berhubungan sehingga muncul gambaran area yang berpotensi menjadi hotel transit.

mapid.co.id/HandsOn2B3



Home Task

GEO MAPID

Code the Map
Decode the Future

Deadline Task: 19 Juni 2026 18.00 WIB

Peserta dapat memilih satu atau beberapa tema use case MAPID berikut:

- **Spatial Plan/Land Development**
- **Retail Expansion/Performance**
- **Agriculture/Environment**
- **Logistics**
- **etc**

Peserta diminta merancang dokumen proyek WebGIS yang menjelaskan:

1. Ide Project
2. Permasalahan yang Diangkat
3. Target Pengguna
4. Data yang Digunakan (termasuk Metodologi)
5. Fitur yang Ingin Dibuat
6. Referensi Tampilan/Inspirasi
7. Output Akhir yang Diharapkan

mapid.co.id/Task2B3

Project Definition : WebGIS Development Bootcamp

Informasi Project

Nama Project WebGIS :

Nama Peserta :

1. Ide Project

Jelaskan secara singkat WebGIS yang ingin dibuat.

.....
.....

2. Permasalahan yang Diangkat

Masalah apa yang ingin dibantu/diselesaikan oleh WebGIS ini?

.....
.....

3. Target Pengguna

Code the Map
Decode the Future

MAPID
Academy

Thank You
So Much!

WebGIS Development Bootcamp